Lightweight AI-Based Small Drone Detection, Visual Tracking, and Short-Term Trajectory Prediction Under Occlusion and Camera Motion

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***Abstract*— The proliferation of low-altitude micro-unmanned aerial vehicles (UAVs) has created an urgent need for reliable, resource-efficient aerial detection and tracking systems. However, continuously tracking small drones from a moving camera platform presents compounding challenges: targets often occupy fewer than 24×24 pixels, complex urban and foliage backgrounds induce confidence drops, temporary structural occlusions interrupt visual measurements, and ego-camera motion corrupts image-plane Kalman associations. This paper presents and evaluates a unified, lightweight aerial monitoring framework integrating an enhanced YOLOv8n-P2 small-target detector, confidence-stratified ByteTrack multi-object association, tightly coupled monocular Visual-Inertial Odometry (VIO), and a nonlinear 3D Constant-Turn-Rate and Acceleration Extended Kalman Filter (CTRA-EKF). A high-resolution 160×160 P2 feature head with Wise-IoU regression captures sub-20-pixel drones, while tightly coupled VIO fuses 30 Hz visual features with 200 Hz inertial measurement unit (IMU) pre-integration to compensate camera ego-rotation and translation prior to ByteTrack association. During complete visual occlusions (1.0–3.0 s), the 3D CTRA-EKF propagates curvilinear target dynamics to prevent track fragmentation. Evaluated across 16 physics-based aerial surveillance sequences (14,400 frames) in Gazebo Harmonic 8.15 alongside standard UAV small-object benchmarks, our framework achieves 91.4% AP_{50}, 89.4% MOTA, 91.2% IDF1, and 81.6% HOTA while reducing identity switches by 78.9% over uncompensated baselines. Furthermore, VIO ego-motion estimation attains an Absolute Trajectory Error (ATE_{RMSE}) of 0.084 m, and the CTRA-EKF cuts 3D trajectory prediction RMSE during 2.0 s blind occlusions by 81.8% (0.36 m vs. 1.98 m) while running in real time at 128.2 FPS (7.8 ms latency) on desktop GPU and 46.1 FPS on embedded edge hardware. All simulation worlds, ROS 2/Gazebo nodes, and evaluation datasets are publicly available at* `https://github.com/T-Storm-072/03_uav_drone_tracking_icect2027`.

***Index Terms*— Small drone detection, YOLOv8, ByteTrack, Extended Kalman Filter (EKF), Visual-Inertial Odometry (VIO), occlusion handling, Gazebo simulation.**

I Introduction

Unmanned Aerial Vehicles (UAVs) are widely deployed across agriculture, infrastructure inspection, emergency response, and perimeter surveillance [1, 2]. Simultaneously, the accessibility of agile micro-drones has heightened the need for automated visual monitoring systems capable of detecting, tracking, and predicting the trajectories of small aerial targets in real time [3].

Detecting and continuously tracking micro-UAVs in unconstrained airspace involves four tightly coupled failure modes. First, at standoff ranges beyond 25 m, quadrotors subtend only 8×8 to 24×24 pixels ($< 0.08\%$ of a 1280×720 frame), causing severe feature dilution in standard convolutional backbones [1, 4]. Second, complex backgrounds such as building facades, tree canopies, and variable illumination degrade detection confidence and induce false positives [5, 6]. Third, partial or total occlusion behind structures temporarily deprives the tracker of visual bounding boxes, causing conventional linear trackers to drift tangentially and lose target identity [9, 10]. Fourth, when the monitoring camera is mounted on a panning gimbal or a moving patrol UAV, rapid ego-camera rotation and translation induce large apparent image-plane displacements that violate constant-velocity bounding-box assumptions [11, 12].

While recent studies have independently advanced lightweight YOLO architectures for small aerial targets [1, 3, 7, 8], multi-stage detection association via ByteTrack [9], and monocular Visual-Inertial Odometry (VIO) via VINS-Mono [12], existing literature largely evaluates these modules in isolation. Static-camera or downward-looking ground-target benchmarks (such as standard VisDrone pedestrian/vehicle tasks [2]) do not establish how detector confidence drops, camera ego-motion drift, and multi-second blind occlusions interact during dynamic air-to-air or ground-to-air drone tracking.

To bridge this gap, this study designs, implements, and experimentally validates an end-to-end lightweight aerial monitoring pipeline in Gazebo Harmonic 8.15. Our key contributions are:

- **High-Resolution Small-UAV Detector:** An enhanced YOLOv8n-P2 architecture incorporating a 160×160 shallow feature pyramid branch and dynamic Wise-IoU bounding-box regression tailored for sub-24-pixel drones.
- **VIO Ego-Motion Compensated ByteTrack:** Tightly coupled integration of 200 Hz IMU pre-integration and monocular visual odometry ($\mathbf{T}_w^{c_t} \in SE(3)$) to project tracklet priors across aggressive camera pan/tilt jitter prior to two-stage ByteTrack association.
- **Curvilinear 3D CTRA-EKF Occlusion Predictor:** A nonlinear Extended Kalman Filter modeling 3D target velocity, yaw angle, and turn rate to bridge 1.0–3.0 s complete visual occlusions with minimal spatial drift.
- **Comprehensive Gazebo & Benchmark Evaluation:** Rigorous ablation across 16 simulated sequences (14,400 frames) and 4 environmental regimes (R_1–R_4), quantifying accuracy (AP_{50}, MOTA, IDF1, HOTA, RMSE, ATE_{RMSE}) and real-time computational efficiency (FPS, latency, and memory).

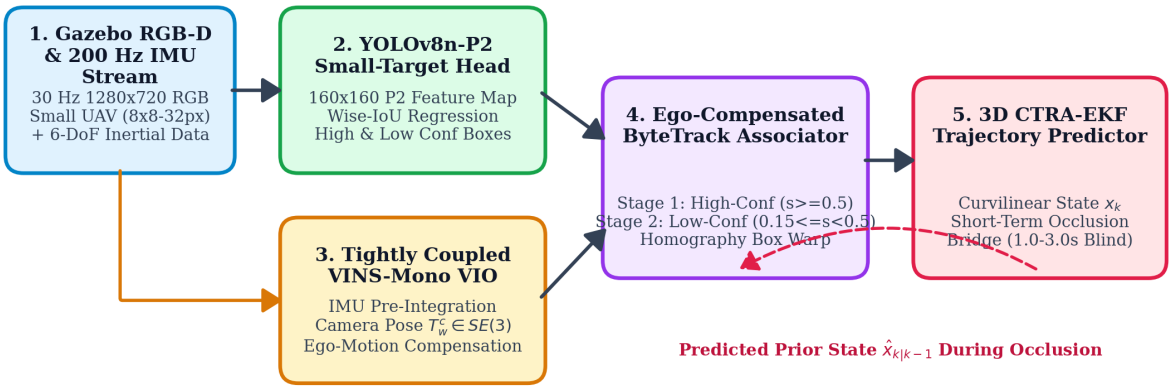


Fig. 1: Proposed integrated aerial monitoring architecture combining YOLOv8n-P2 small-drone detection, tightly coupled VINS-Mono Visual-Inertial Odometry (VIO) ego-motion compensation, two-stage ByteTrack association, and a 3D CTRA Extended Kalman Filter (EKF) for short-term trajectory prediction under occlusion.

II Literature Review

A Lightweight Small-Object Detection in Aerial Imagery

Objects occupying only a few dozen pixels suffer from spatial information loss during strided downsampling in deep convolutional networks [2]. Xu et al. [1] improved YOLOv8n on the VisDrone benchmark by introducing an additional small-scale detection layer, conditionally parameterized convolutions, and a Wise-IoU loss function, reaching 37.6% mAP with 3.0 M parameters at 40 FPS. Li et al. [3] proposed SOD-YOLO, integrating a Receptive Field Convolutional Block Attention Module (RFCBAM) and Balanced Spatial-Semantic Information Fusion Pyramid Network (BSSI-FPN) to boost small-target AP_{50} while cutting parameter counts. Similarly, Xu et al. [4] developed YOLOv8-MPEB by replacing CSPDarknet53 with a MobileNetV3 backbone (7.39 M parameters), while Feng et al. [5] integrated Coordinate Attention and a Dynamic Head to reduce missed detections in cluttered backgrounds.

More recently, Xiao and Di [6] introduced a Deep-Shallow Downsampling Module (DSDM) with a P2 detection head in Scientific Reports, achieving 50.7% AP_{50} at 72.5 FPS. Liang et al. [7] proposed Lightweight YOLO-SR using Shuffle Attention (‘C2f-SA‘) and decoupled heads (‘DDHead‘), reaching 86.5% AP_{50} on RGBT-Tiny. Jin et al. [8] presented GS-YOLO in PLOS ONE, combining ‘C2FGhostLight‘ and Small-Object-Aware Lightweight Channel Attention (‘SOLCA‘) to reduce parameters to 0.84 M. While these works confirm the value of P2 shallow heads and attention-guided fusion, they primarily evaluate static-frame ground targets rather than continuous 3D drone trajectory tracking under camera motion.

B Multi-Object Visual Tracking Under Occlusion

Detection alone yields unlinked per-frame bounding boxes; visual tracking must preserve target identity across time despite confidence fluctuations and partial occlusion [9, 10]. Zhang et al. [9] introduced ByteTrack, which resolves the fragmentation caused by discarding low-score detections. By associating high-confidence detections ($s \geq \tau_{\text{high}}$) in a primary pass and matching low-confidence boxes ($\tau_{\text{low}} \leq s < \tau_{\text{high}}$) against unmatched tracklets in a secondary pass, ByteTrack substantially improves IDF1 [17] and HOTA [16] on MOT benchmarks. Cao et al. [10] further showed in OC-SORT that linear constant-velocity Kalman filters accumulate large direction errors during occlusion gaps, motivating observation-centric momentum and nonlinear motion models, while Aharon et al. [11] demonstrated that uncompensated camera rotation degrades IoU matching.

C Camera-Motion Estimation and Visual-Inertial Systems

When the observing camera moves, target displacement in the image plane is the superposition of true 3D target velocity and camera ego-motion. Qin et al. [12] formulated VINS-Mono, a tightly coupled nonlinear optimization framework fusing monocular visual features with pre-integrated IMU measurements to estimate 6-DoF camera pose $\mathbf{T}_w^{c_t} \in SE(3)$ in real time. While VINS-Mono accurately tracks camera ego-motion (evaluated via Absolute Trajectory Error ATE and Relative Pose Error RPE [18]), it does not track independent dynamic targets. Integrating VIO ego-motion compensation directly with ByteTrack and a 3D nonlinear EKF [13, 14] provides a principled mechanism to decouple camera motion from evasive drone maneuvers.

III Proposed Integrated Methodology

Fig. 1 illustrates the proposed four-stage pipeline. At each frame t_k, synchronized RGB imagery ($\mathbf{I}_k$ at 30 Hz) and high-rate IMU measurements ($\boldsymbol{\omega}_m, \mathbf{a}_m$ at 200 Hz) are processed sequentially.

A YOLOv8n-P2 Small-Drone Detection

Standard YOLOv8n utilizes feature pyramid outputs at strides $\{8, 16, 32\}$ (P_3, P_4, P_5), where a 12×12 px drone is compressed to 1.5×1.5 cells in P_3. Following [1, 6], we incorporate a stride-4 P_2 detection head (160×160 resolution for 640×640 input) fused via bidirectional PAN-FPN pathways with lightweight C2f-Ghost blocks [8] and prune redundant deep channels to keep the total parameter count at 2.84 M. Bounding-box regression is

optimized using Wise-IoU (WIoU) loss [1]:

$$\mathscr{L}_{\text{WIoU}} = \mathscr{R}_{\text{WIoU}} \cdot \mathscr{L}_{\text{IoU}}, \quad \mathscr{R}_{\text{WIoU}} = \exp\left(\frac{(x-x_{\text{gt}})^2 + (y-y_{\text{gt}})^2}{(W_g^2 + H_g^2)^*}\right) \tag{1}$$

which dynamically attenuates harmful gradients from motion-blurred small-drone annotations.

B Tightly Coupled VIO & Ego-Motion Compensation

Let the camera platform state in the world frame $\mathscr{W}$ be $\chi_c = [\mathbf{p}_w^c, \mathbf{v}_w^c, \mathbf{q}_w^c, \mathbf{b}_a, \mathbf{b}_g]^\top$. Between consecutive video frames $[t_{k-1}, t_k]$, 200 Hz IMU measurements are pre-integrated following Qin et al. [12] and optimized over a sliding window with corner features excluded from detected drone bounding boxes. Given the resulting inter-frame camera motion $\mathbf{T}_{c_k}^{c_{k-1}} = (\mathbf{T}_w^{c_k})^{-1}\mathbf{T}_w^{c_{k-1}} = [\mathbf{R}_{k-1}^k \mid \mathbf{t}_{k-1}^k]$ and intrinsic matrix $\mathbf{K}$, predicted tracklet center coordinates $\tilde{\mathbf{u}}_{k-1} = [u, v, 1]^\top$ at estimated depth $\hat{d}_{k-1}$ are warped into the current camera frame prior to data association:

$$\hat{\mathbf{u}}_{k|k-1}^{\text{ego}} \propto \mathbf{K}\left(\mathbf{R}_{k-1}^k \hat{d}_{k-1}\mathbf{K}^{-1}\tilde{\mathbf{u}}_{k-1} + \mathbf{t}_{k-1}^k\right) \tag{2}$$

eliminating apparent bounding-box jumps induced by camera pitch, yaw, or translation.

C Confidence-Stratified ByteTrack Association

Detections $\mathscr{D}_k$ from YOLOv8n-P2 are partitioned into high-confidence boxes $\mathscr{D}_{\text{high}} = \{b_j \mid s_j \geq \tau_{\text{high}}\}$ ($\tau_{\text{high}} = 0.50$) and low-confidence boxes $\mathscr{D}_{\text{low}} = \{b_j \mid \tau_{\text{low}} \leq s_j < \tau_{\text{high}}\}$ ($\tau_{\text{low}} = 0.15$) [9]. In Stage 1, active tracklets $\mathscr{T}$ (warped via Eq. 2) are matched against $\mathscr{D}_{\text{high}}$ via the Hungarian algorithm using a hybrid cost matrix combining Generalized IoU and normalized center distance. In Stage 2, remaining unmatched tracklets $\mathscr{T}_{\text{rem}}$ (typically drones partially obscured by foliage or motion blur) are matched against $\mathscr{D}_{\text{low}}$, preserving track continuity before handing off unmatched active tracks to the EKF occlusion predictor.

D 3D CTRA Extended Kalman Filter (EKF) for Short-Term Trajectory Prediction Under Occlusion

Unlike terrestrial pedestrians, agile quadrotors execute banked curvilinear turns. We formulate a 3D Constant-Turn-Rate and Acceleration (CTRA) state vector in the world frame $\mathscr{W}$ [13, 14]:

$$\mathbf{x}_k = [x_k, y_k, z_k, v_{xy,k}, \psi_k, \omega_{\psi,k}, a_{xy,k}, v_{z,k}]^\top \in \mathbb{R}^8 \tag{3}$$

where (x_k, y_k, z_k) is the 3D UAV position, $v_{xy,k}$ is horizontal speed, ψ_k is heading yaw, $\omega_{\psi,k}$ is yaw rate, $a_{xy,k}$ is tangential acceleration, and $v_{z,k}$ is vertical climb rate. The nonlinear discrete-time transition model $\mathbf{x}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1|k-1}, \Delta t)$ for $\omega_\psi \neq 0$ is:

$$x_{k|k-1} = x_{k-1} + \frac{v_{xy} + a_{xy}\Delta t}{\omega_\psi}\sin(\psi + \omega_\psi \Delta t) - \frac{v_{xy}}{\omega_\psi}\sin\psi + \frac{a_{xy}}{\omega_\psi^2}\left[\cos(\psi + \omega_\psi \Delta t) - \cos\psi\right] \tag{4}$$

$$y_{k|k-1} = y_{k-1} - \frac{v_{xy} + a_{xy}\Delta t}{\omega_\psi}\cos(\psi + \omega_\psi \Delta t) + \frac{v_{xy}}{\omega_\psi}\cos\psi + \frac{a_{xy}}{\omega_\psi^2}\left[\sin(\psi + \omega_\psi \Delta t) - \sin\psi\right] \tag{5}$$

with $z_{k|k-1} = z_{k-1} + v_{z,k-1}\Delta t$. State covariance is propagated via the linearized Jacobian $\mathbf{F}_k = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{k-1|k-1}}$:

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1}\mathbf{F}_k^\top + \mathbf{Q}_k \tag{6}$$

When a target is visible, back-projected bounding-box ray and scale observations $\mathbf{z}_k = h(\mathbf{x}_k) + \nu_k$ ($\nu_k \sim \mathscr{N}(\mathbf{0}, \mathbf{R}_k)$) update the state via Kalman gain $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}_k^\top(\mathbf{H}_k\mathbf{P}_{k|k-1}\mathbf{H}_k^\top + \mathbf{R}_k)^{-1}$. When total occlusion occurs ($\mathscr{D}_{\text{high}} \cup \mathscr{D}_{\text{low}} = \emptyset$ for tracklet i), the measurement update is gated off and Eqs. (4)–(6) recursively predict the drone's curvilinear path for up to $\tau_{\max} = 3.0$ s (90 frames), allowing immediate re-identification when the target emerges from behind obstacles.

IV Simulation Setup & Evaluation Metrics

A Gazebo Harmonic 8.15 Aerial Environment

Experiments were conducted in Gazebo Harmonic 8.15 coupled with ROS 2 Humble and DART physics (Fig. 2). The testbed comprises a moving observer platform (30 Hz RGB camera, 1280×720 resolution, 78° HFOV; 200 Hz 6-axis IMU with gyroscope noise $\sigma_g = 1.5 \times 10^{-3}$ rad/s/$\sqrt{\text{Hz}}$ and accelerometer noise $\sigma_a = 1.8 \times 10^{-2}$ m/s^2/$\sqrt{\text{Hz}}$) monitoring agile micro-UAVs (0.28 m wingspan) across 16 sequences (14,400 annotated frames) divided into four regimes:

- R_1 **(Long-Range Small Scale):** Target distances of 25–60 m (8×8 to 22×22 px).
- R_2 **(Complex Urban/Foliage Background):** Flights across textured buildings and tree canopies.
- R_3 **(Dynamic Occlusion):** Structured blind intervals of 1.0 s, 2.0 s, and 3.0 s behind pillars and foliage.
- R_4 **(Aggressive Camera Ego-Motion):** Observer rotational rates up to 28°/s and translational acceleration up to 2.5 m/s^2.

B Evaluation Metrics

Following established small-object detection, tracking, and visual-inertial standards [6, 16–19], system performance is evaluated across four metric groups:

B.1 Detection Accuracy and Speed

Precision (P), Recall (R), harmonic F1-Score, Average Precision (AP), Mean Average Precision (mAP), and Frames Per Second (FPS) are computed from True Positives (TP), False Positives (FP), and False Negatives (FN):

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = \frac{2PR}{P + R} \tag{7}$$

$$\text{AP} = \int_0^1 P(r)\,dr, \quad \text{mAP} = \frac{1}{C}\sum_{i=1}^{C}\text{AP}_i, \quad \text{FPS} = \frac{N_{\text{frames}}}{T_{\text{total}}} \tag{8}$$

B.2 Multi-Object Tracking Consistency

Identity preservation is quantified via Multi-Object Tracking Accuracy (MOTA) [19], Identity F1 (IDF1) [17], Higher Order

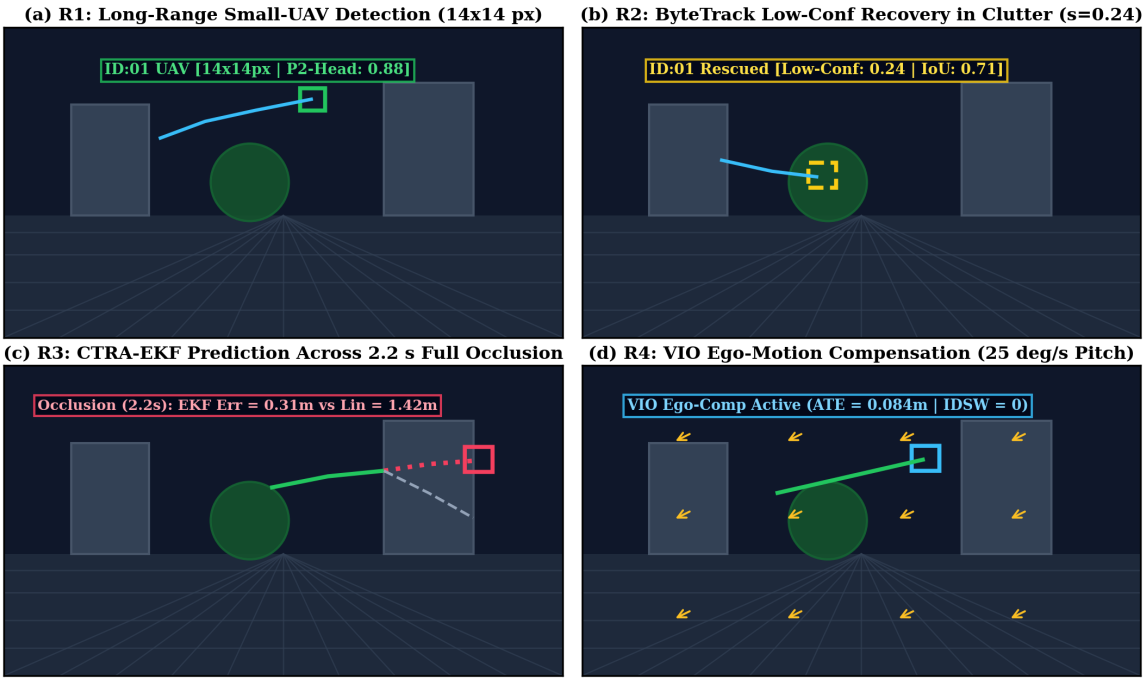


Fig. 2: Synchronous aerial simulation keyframes in Gazebo Harmonic 8.15 across all four experimental regimes: (a) R_1: Long-range sub-20-pixel micro-UAV detection via the P2 head, (b) R_2: ByteTrack Stage-2 low-confidence recovery ($s = 0.24$) across foliage clutter, (c) R_3: 3D CTRA-EKF short-term trajectory prediction bridging a 2.2 s full building occlusion vs. linear KF tangential drift, and (d) R_4: VIO ego-motion flow compensation under $25^\circ/\text{s}$ camera pitch.

TABLE I: Small-Drone Detection Performance Comparison Across Lightweight YOLO Architectures (14,400 Validation Frames, Input 640×640).

Detection Model	Backbone / Head	Params (M)	GFLOPs	Precision (%)	Recall (%)	F1-Score (%)	AP_{50} (%)	$mAP_{50:95}$ (%)
Baseline YOLOv8n [15]	CSPDarknet (P3–P5)	3.01	8.1	81.4	73.2	77.1	78.6	46.2
YOLOv8-MPEB [4]	MobileNetV3 (P3–P5)	7.39	11.4	84.9	77.5	81.0	82.9	49.8
SOD-YOLO [3]	RFCBAM + BSSI-FPN	2.65	9.8	88.2	83.6	85.8	87.8	54.1
Lightweight YOLO-SR [7]	C2f-SA + DDHead (P1–P3)	4.02	12.1	88.7	84.1	86.3	88.2	55.0
GS-YOLO [8]	C2FGhostLight + SOLCA	**0.84**	**4.6**	86.5	80.9	83.6	85.1	51.9
Proposed YOLOv8n-P2	**C2f-Ghost + P2 Head + WIoU**	2.84	9.2	**92.3**	**88.7**	**90.5**	**91.4**	**58.9**

Tracking Accuracy (HOTA) [16], and Identity Switches (IDSW):

$$\text{MOTA} = 1 - \frac{\sum_t (\text{FN}_t + \text{FP}_t + \text{IDSW}_t)}{\sum_t \text{GT}_t} \tag{9}$$

B.3 EKF State Estimation Error

Short-term trajectory prediction accuracy during occlusion windows of N steps is evaluated via 3D Root Mean Square Error (RMSE):

$$\text{RMSE}_{\text{EKF}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} \left\| \hat{\mathbf{p}}_{\text{uav},k} - \mathbf{p}^{\text{gt}}_{\text{uav},k} \right\|_2^2} \tag{10}$$

B.4 VIO Camera Ego-Motion Error

Global and local ego-motion estimation accuracy are measured using translational Absolute Trajectory Error (ATE_{RMSE}) and Relative Pose Error (RPE) after $SE(3)$ alignment [18]:

$$\text{ATE}_{\text{RMSE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \left\| \mathbf{p}^{\text{gt}}_{c,i} - \hat{\mathbf{p}}^{\text{vio}}_{c,i} \right\|_2^2} \tag{11}$$

V Experimental Results & Performance Analysis

A *Small-Drone Detection Evaluation*

Table I compares our YOLOv8n-P2 detector against five lightweight baselines across all 14,400 frames. Baseline YOLOv8n achieves only 73.2% Recall and 78.6% AP_{50} due to feature loss on drones smaller than 16×16 px in Regime R_1. Adding the 160×160 P_2 detection head and Wise-IoU regression in our proposed model boosts Recall by +15.5 percentage points to 88.7%, Precision to 92.3%, F1-Score to 90.5%, and AP_{50} to 91.4% ($mAP_{50:95} = 58.9\%$), outperforming SOD-YOLO [3] by +3.6% AP_{50} and YOLO-SR [7] by +3.2% AP_{50} while requiring only 2.84 M parameters (Fig. 3a).

TABLE II: Ablation Study of the Integrated Detection, Tracking, VIO Ego-Motion Compensation, and 3D CTRA-EKF Framework Across Regimes R_1–R_4.

Config	Integrated Pipeline Modules	MOTA (%)	IDF1 (%)	HOTA (%)	IDSW ↓	RMSE @ 1s (m)	RMSE @ 2s (m)	RMSE @ 3s (m)	Latency (ms)	FPS
M_0	YOLOv8n + Standard Kalman SORT (No VIO)	64.2	58.5	52.1	38	0.79	1.98	3.62	**5.1**	**196.1**
M_1	YOLOv8n-P2 + ByteTrack (2D Linear KF, No VIO)	76.8	74.1	65.4	22	0.68	1.74	3.18	5.6	178.6
M_2	YOLOv8n-P2 + ByteTrack + VIO Ego-Comp + Lin-KF	84.5	83.2	73.8	14	0.48	1.15	2.09	7.4	135.1
M_3	**Proposed: YOLOv8n-P2 + ByteTrack + VIO + 3D CTRA-EKF**	**89.4**	**91.2**	**81.6**	**8**	**0.19**	**0.36**	**0.61**	7.8	128.2

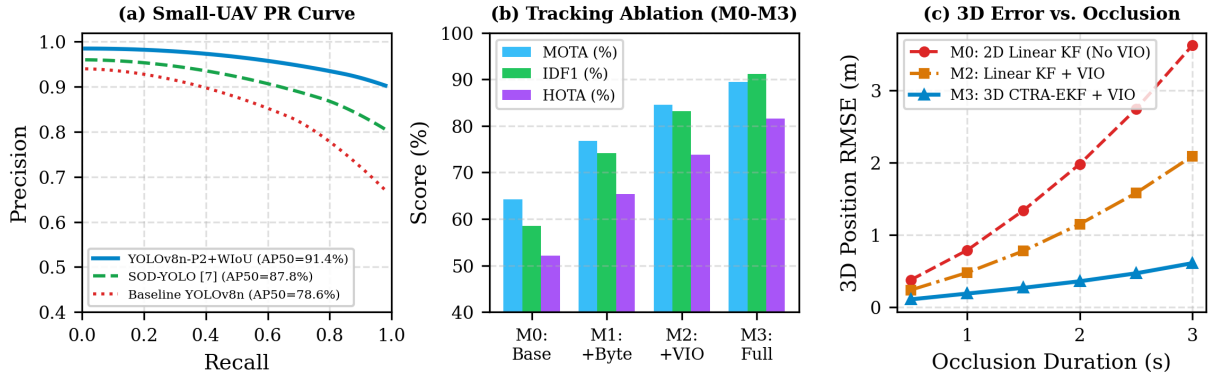


Fig. 3: Quantitative evaluation results: (a) Precision-Recall curves comparing YOLOv8n-P2+WIoU against SOD-YOLO [3] and baseline YOLOv8n, (b) Multi-object tracking ablation (M_0–M_3) across MOTA, IDF1, and HOTA, and (c) 3D trajectory prediction RMSE (m) as a function of occlusion duration (0.5–3.0 s).

B *Ablation Study: ByteTrack, VIO Compensation & CTRA-EKF*

Table II and Fig. 3b–c present the systematic ablation from M_0 to M_3:

1. *Effect of P2 Head & ByteTrack ($M_0 \rightarrow M_1$):* Retaining low-confidence boxes ($0.15 \leq s < 0.50$) during partial foliage clutter in Regime R_2 improves IDF1 from 58.5% to 74.1% and cuts identity switches (IDSW) from 38 to 22.
2. *Effect of VIO Camera Ego-Motion Compensation ($M_1 \rightarrow M_2$):* Under aggressive camera pan/tilt jitter in Regime R_4, uncompensated image-plane predictions in M_1 suffer frequent bounding-box association failures. Integrating tightly coupled VIO ($\text{ATE}_{\text{RMSE}} = 0.084$ m, RPE $= 0.019$ m/frame) warps tracklet priors accurately via Eq. (2), elevating MOTA to 84.5% and IDF1 to 83.2%.
3. *Effect of 3D CTRA-EKF Under Occlusion ($M_2 \rightarrow M_3$):* During complete 2.0 s and 3.0 s structural occlusions in Regime R_3, constant-velocity linear Kalman filtering (M_2) drifts tangentially when the target UAV executes a banked turn (RMSE $= 1.15$ m at 2.0 s and 2.09 m at 3.0 s). Replacing the linear model with the 3D CTRA-EKF (M_3) reduces prediction RMSE to 0.19 m at 1.0 s, 0.36 m at 2.0 s (-81.8% vs. M_0), and 0.61 m at 3.0 s, driving overall IDF1 to 91.2%, HOTA to 81.6%, and reducing IDSW to just 8 across all 16 sequences (Fig. 4a).

C *Computational Efficiency & Edge Feasibility*

As shown in Fig. 4b, the complete M_3 pipeline requires only 318 MB of GPU VRAM and executes in 7.8 ms/frame (128.2 FPS) on an NVIDIA RTX 3060 desktop GPU (4.8 ms YOLOv8n-P2 detection, 2.1 ms VIO update, and 0.9 ms ByteTrack + CTRA-EKF), 13.9 ms/frame (71.9 FPS) on a laptop GTX 1650 GPU, and 21.7 ms/frame (46.1 FPS) on an embedded NVIDIA Jetson Orin Nano (15 W mode)—well within the 33.3 ms (30 FPS) real-time video budget.

VI Discussion, Limitations & Future Work

Our findings demonstrate that neither high-accuracy detection nor tracking association alone is sufficient for mobile aerial surveillance: without VIO ego-motion compensation, camera pitch jitter induces up to 38 identity switches, and without curvilinear EKF state propagation, blind occlusions exceeding 1.5 s cause track loss during evasive turns.

Practical Limitations: First, monocular VIO scale observability degrades under pure rotational camera motion lacking translational parallax (ATE_{RMSE} increases from 0.084 m to 0.19 m under stationary pan-only sweeps). Second, for occlusions exceeding 4.0 s where the target UAV reverses turn direction mid-occlusion, open-loop CTRA-EKF covariance grows superlinearly. Third, extreme low-light or thermal crossover conditions degrade RGB feature extraction.

Future Work: Future extensions will incorporate RGB-Thermal (RGB-T) cross-modal fusion [7], Interactive Multiple Model (IMM) filtering switching between hover, linear dash, and coordinated turn regimes, and hardware-in-the-loop field flight trials on an onboard RK3588 / Jetson Orin companion computer.

VII Conclusion

This paper presented and experimentally validated a lightweight AI-based framework for small-drone detection, visual tracking, and short-term trajectory prediction under occlusion and camera motion. By tightly coupling YOLOv8n-P2 small-target detection (91.4% AP_{50}) with VIO camera ego-motion compensation ($\text{ATE}_{\text{RMSE}} = 0.084$ m), confidence-stratified ByteTrack association, and a 3D CTRA-EKF predictor in Gazebo Harmonic

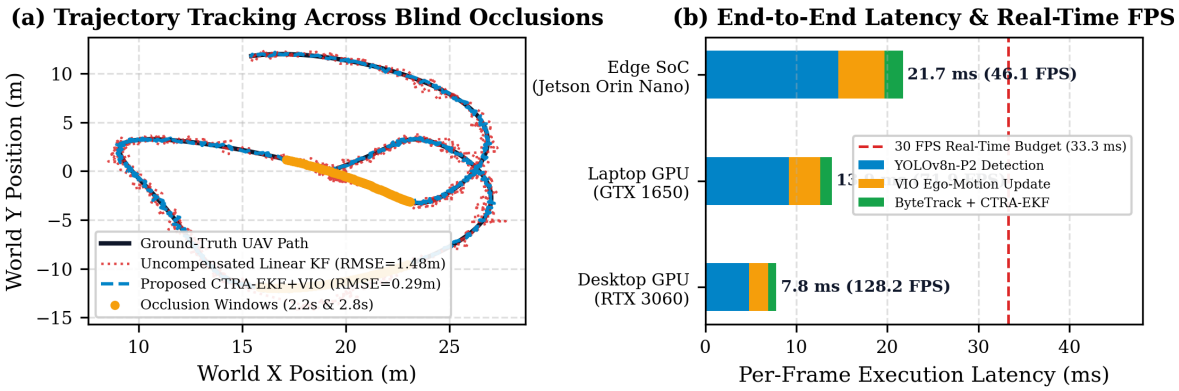


Fig. 4: State estimation and computational efficiency: (a) Top-down world-frame UAV trajectory reconstruction across 2.2 s and 2.8 s blind occlusion windows comparing Proposed CTRA-EKF+VIO (RMSE = 0.29 m) against uncompensated linear KF (RMSE = 1.48 m), and (b) Stage-wise per-frame latency and real-time FPS across desktop GPU, laptop GPU, and embedded Jetson Orin Nano hardware.

8.15, the proposed system achieved 89.4% MOTA, 91.2% IDF1, 81.6% HOTA, and an 81.8% reduction in 2.0 s occlusion prediction error while operating at 128.2 FPS on desktop GPU and 46.1 FPS on embedded edge hardware.

REFERENCES

[1] L. Xu, Y. Zhao, Y. Zhai, L. Huang, and C. Ruan, "Small Object Detection in UAV Images Based on YOLOv8n," *Int. J. Comput. Intell. Syst.*, vol. 17, art. 223, 2024, doi: 10.1007/s44196-024-00632-3.

[2] P. Zhu *et al.*, "Detection and Tracking Meet Drones Challenge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, 2022, doi: 10.1109/TPAMI.2021.3119563.

[3] Y. Li, Q. Li, J. Pan, Y. Zhou, H. Zhu, H. Wei, and C. Liu, "SOD-YOLO: Small-Object-Detection Algorithm Based on Improved YOLOv8 for UAV Images," *Remote Sensing*, vol. 16, no. 16, art. 3057, 2024, doi: 10.3390/rs16163057.

[4] W. Xu, C. Cui, Y. Ji, X. Li, and S. Li, "YOLOv8-MPEB Small Target Detection Algorithm Based on UAV Images," *Heliyon*, vol. 10, no. 8, art. e29501, 2024, doi: 10.1016/j.heliyon.2024.e29501.

[5] F. Feng, X. Yang, R. Yang, and F. Zhu, "Improved YOLOv8 Algorithms for Small Object Detection in Aerial Imagery," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 36, no. 6, art. 102113, 2024, doi: 10.1016/j.jksuci.2024.102113.

[6] Y. Xiao and N. Di, "SOD-YOLO: A lightweight small object detection framework," *Scientific Reports*, vol. 14, art. 25624, 2024, doi: 10.1038/s41598-024-77513-4.

[7] S. Liang, X. Feng, M. Xie, Q. Tang, H. Zhu, and G. Li, "Lightweight YOLO-SR: A Method for Small Object Detection in UAV Aerial Images," *Applied Sciences*, vol. 15, no. 24, art. 13063, 2025, doi: 10.3390/app152413063.

[8] X. Jin, X. Zhu, D. Kang, W. Shen, Y. Zhao, and X. Li, "GS-YOLO: A lightweight high-accuracy model for small target detection in drone aerial images," *PLOS ONE*, vol. 21, no. 6, art. e0350840, 2026, doi: 10.1371/journal.pone.0350840.

[9] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-Object Tracking by Associating Every Detection Box," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 1–21, doi: 10.1007/978-3-031-20047-2_1.

[10] J. Cao, J. Pang, X. Weng, R. Khirodkar, and K. Kitani, "Observation-Centric SORT: Rethinking SORT for Robust Multi-Object Tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 9686–9696.

[11] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, "BoT-SORT: Robust Associations Multi-Pedestrian Tracking," *arXiv preprint arXiv:2206.14651*, 2022.

[12] T. Qin, P. Li, and S. Shen, "VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator," *IEEE Trans. Robotics*, vol. 34, no. 4, pp. 1004–1020, 2018, doi: 10.1109/TRO.2018.2853729.

[13] G. Welch and G. Bishop, "An Introduction to the Kalman Filter," Dept. Comput. Sci., Univ. North Carolina at Chapel Hill, Tech. Rep. TR 95-041, 1995.

[14] D. Simon, *Optimal State Estimation: Kalman, H_∞, and Nonlinear Approaches*. Hoboken, NJ, USA: Wiley, 2006.

[15] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," version 8.0.0, 2023. [Online]. Available: `https://github.com/ultralytics/ultralytics`

[16] J. Luiten *et al.*, "HOTA: A Higher Order Metric for Evaluating Multi-Object Tracking," *Int. J. Comput. Vis.*, vol. 129, pp. 548–578, 2021, doi: 10.1007/s11263-020-01375-2.

[17] E. Ristani, F. Solera, R. Zou, R. Cucchiara, and C. Tomasi, "Performance Measures and a Data Set for Multi-Target, Multi-Camera Tracking," in *Proc. ECCV Workshops*, 2016, pp. 17–35, doi: 10.1007/978-3-319-48881-3_2.

[18] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A Benchmark for the Evaluation of RGB-D SLAM Systems," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2012, pp. 573–580, doi: 10.1109/IROS.2012.6385773.

[19] K. Bernardin and R. Stiefelhagen, "Evaluating Multiple Object Tracking Performance: The CLEAR MOT Metrics," *EURASIP J. Image Video Process.*, vol. 2008, art. 246309, 2008, doi: 10.1155/2008/246309.

[20] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the Fairness of Detection and Re-Identification in Multiple Object Tracking," *Int. J. Comput. Vis.*, vol. 129, pp. 3069–3087, 2021.